Our project aims to engineer BtSULT to catalyze the sulfonation of lithocholic acid (LCA), potentially improving bile acid metabolism in individuals experiencing debilitating post-cholecystectomy syndrome (PCS). The goal of our in vitro trials was to express, quantify, purify, and evaluate the structural and functional properties of our target sulfotransferase enzyme.

The experimental workflow encompassed plasmid transformation, colony screening, protein expression, purification, and biochemical evaluation. Plasmid constructs were transformed into competent Escherichia coli (*E.Coli*) BL21 cells and screened using **Colony Polymerase Chain Reaction (PCR)** to identify colonies containing the desired construct. Positive colonies were subsequently cultured to establish larger-scale expression cultures:

> *Plasmid Vector and Insert Details: pET_28(+), Kan. R, T7 Primer, lacI operon*

Protein concentration was quantified using the **Bicinchoninic Acid (BCA) Assay**, while protein composition and apparent molecular weight were evaluated using Coomassie Brilliant Blue staining following sodium dodecyl sulfate–**Polyacrylamide Gel Electrophoresis (SDS-PAGE)**.

> *Avg. Purified Sample Concentration (under non-conflicting growth conditions):*
>
> *238.759µg/mL (σ = 32.558µg/mL) (Std. Curve $R^2 \cong 0.95$)*

Protein expression was induced using isopropyl β-D-1-thiogalactopyranoside (IPTG). Skim milk was also investigated as an alternative or supplementary expression-induction strategy deriving from necessity. Following expression, crude intracellular materials were harvested and lysed to recover the protein-containing fraction. The target protein was then isolated through His-tag affinity purification.

> *Cell lysis via B-PER™ Reagent(Non-ionic 20 mM Tris•HCl buffer at pH 7.5), Protein Purification via His-Spin Protein Miniprep Nickel-charged His-Affinity Gel [IMAC] (Zymo P2002)*